MATH 350-2 Advanced Calculus

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Outline

1. Real Analysis Lecture 21
   - Series tests
   - Infinite products

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Tests for Convergence

Theorem (Comparison Test)

If $a_n > 0$ and $b_n > 0$ and there exists $c > 0$ and $N \in \mathbb{Z}_+$ with $a_n < cb_n$ for all $n \geq N$ then

$$\sum b_n \text{ converges} \quad \Rightarrow \quad \sum a_n \text{ converges}$$

Proof.

Suppose □

Tests for Convergence

Theorem (Limit Comparison Test)

If $a_n > 0$ *and* $b_n > 0$ *and suppose that* $\lim a_n/b_n = c > 0$. *Then*

$$\sum b_n \text{ converges} \quad \Leftrightarrow \quad \sum a_n \text{ converges}$$

Limit infimum/limit supremum

Definition

Let a_n be a sequence. Then

$$\liminf a_n = \lim_n \inf\{a_k : k \geq n\}$$

$$\limsup a_n = \lim_n \sup\{a_k : k \geq n\}$$

Theorem

If a_n is bounded, then $\limsup a_n$ *and* $\liminf a_n$ *exist.*

More generally, we write $\liminf a_n = \pm\infty$ and $\limsup a_n = \pm\infty$.

Tests for Convergence

Theorem (Ratio Test)

Let $r = \liminf |a_{n+1}/a_n|$ *and* $R = \limsup |a_{n+1}/a_n|$.

(a) *if* $R < 1$ *then the series is absolutely convergent*

(b) *if* $r > 1$ *then the series is divergent*

(c) *if* $r \leq 1 \leq R$ *the test is inconclusive*

Tests for Convergence

Theorem (Root Test)

Let $\rho = \limsup \sqrt[n]{|a_n|}$.

(a) *if* $\rho < 1$ *then the series is absolutely convergent*

(b) *if* $\rho > 1$ *then the series is divergent*

(c) *if* $\rho = 1$ *the test is inconclusive*

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Basic definition

Definition

An **infinite product** is an ordererd pair of sequences $(\{a_n\}, \{p_n\})$ where

$$p_n = a_1 a_2 \dots a_n.$$

The sequence $\{p_n\}$ is called the **sequence of partial products** and p_n is called the n'th partial product. We say the series converges or diverges if $\{p_n\}$ converges or diverges.

Notation:

$$a_1 a_2 a_3 \dots \quad \text{or} \quad \prod_{n=1}^{\infty} a_n.$$

Basic definition

We write

$$\prod_{n=1}^{\infty} a_n = L$$

to mean that the limit of the sequence p_n is L. We call L the **product** of the series.

Examples

$$\prod_{n=1}^{\infty}\left(1+\frac{1}{2^{2^n}}\right)$$

$$p_n=\left(1+\frac{1}{2}\right)\left(1+\frac{1}{4}\right)\left(1+\frac{1}{16}\right)\cdots\left(1+\frac{1}{2^{2^n}}\right)$$

Can we compute p_n?

Examples

$$\begin{aligned}\left(1-\frac{1}{2}\right)p_n &= \left(1-\frac{1}{2}\right)\left(1+\frac{1}{2}\right)\left(1+\frac{1}{4}\right)\left(1+\frac{1}{16}\right)\cdots\left(1+\frac{1}{2^n}\right)\\ &= \left(1-\frac{1}{4}\right)\left(1+\frac{1}{4}\right)\left(1+\frac{1}{16}\right)\cdots\left(1+\frac{1}{2^n}\right)\\ &= \left(1-\frac{1}{16}\right)\left(1+\frac{1}{16}\right)\cdots\left(1+\frac{1}{2^n}\right)\\ &= 1-\frac{1}{2^{2^n}} \quad\Longrightarrow\quad p_n = 2-\frac{1}{2^{2^n-1}}\end{aligned}$$

$$\prod_{n=1}^{\infty}\left(1+\frac{1}{2^{2^n}}\right) = \lim_{n\to\infty} p_n = 2$$

Examples

Weierstrass product of sine

$$\sin(\pi x) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right)$$

- first thought of by Euler.
- can be used to prove

$$\sum \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Examples

Let $q_1 = 2, q_2 = 3, q_3 = 5, \ldots$ be the prime integers.
The infinite product

$$\zeta(s) = \left(\frac{1}{1-2^{-s}}\right)\left(\frac{1}{1-3^{-s}}\right)\left(\frac{1}{1-5^{-s}}\right)\left(\frac{1}{1-7^{-s}}\right)\left(\frac{1}{1-11^{-s}}\right)\cdots$$

$$\zeta(s) = \prod_{n=1}^{\infty} \frac{1}{1-q_n^{-s}}$$

converges for $s > 1$.
This product is called the **Riemann zeta function**.

Riemann zeta

There's an amazing way to convert to a series:

Theorem

For any real number $s > 1$

$$\zeta(s) = \prod_{n=1}^{\infty} \frac{1}{1 - q_n^{-s}} = \sum_{m=1}^{\infty} \frac{1}{m^s}$$

Riemann zeta

Here's why:

$$\frac{1}{2^s}\sum_{m=1}^{\infty}\frac{1}{m^s}=\sum_{m=1}^{\infty}\frac{1}{(2m)^s}$$

$$\left(1-\frac{1}{2^s}\right)\sum_{m=1}^{\infty}\frac{1}{m^s}=\sum_{2\nmid m}^{\infty}\frac{1}{m^s}$$

$$\left(1-\frac{1}{2^s}\right)\left(1-\frac{1}{3^s}\right)\sum_{m=1}^{\infty}\frac{1}{m^s}=\sum_{2,3\nmid m}^{\infty}\frac{1}{m^s}$$

$$\left[\prod_{k=1}^{n}\left(1-\frac{1}{p_k^s}\right)\right]\sum_{m=1}^{\infty}\frac{1}{m^s}=\sum_{p_2,p_3,\ldots,p_n\nmid m}^{\infty}\frac{1}{m^s}$$

Riemann zeta

Taking the limit as $n \to \infty$:

$$\zeta(s)^{-1} \sum_{m=1}^{\infty} \frac{1}{m^s} = \sum_{m \text{ not divisible by any prime}} \frac{1}{m^s} = 1.$$

$$\zeta(s) = \sum_{m=1}^{\infty} \frac{1}{m^s}.$$

Examples

Theorem

For any real number $s > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Therefore

$$\frac{1}{(1-1/2^2)(1-1/3^2)(1-1/5^2)(1-1/7^2)(1-1/11^2)\ldots} = \frac{\pi^2}{6}.$$

Relation to series

Theorem

Assume $a_n > 0$ for all n. Then $\prod(1 + a_n)$ converges if and only if $\sum a_n$ converges.

Relation to series

Definition

We say that a product $\prod(1 + a_n)$ **converges absolutely** if $\prod(1 + |a_n|)$ converges.

Theorem

If $\prod(1 + a_n)$ converges absolutely (equiv. $\sum a_n$ converges absolutely), then $\prod(1 + a_n)$ converges.